# OMEX Combination Report — SLS_Concentration=250.0

vcornea_sls_sweep_combo001 (point 1 of 3) | 2 replicate archive(s) | vivarium_wrapper 1.0.0

## Combination Overview

| | |
|---|---|
| **Scan point** | SLS_Concentration=250.0 (point 1 of 3) |
| **Simulation** | SimTime 30 days (= 7200 MCS = 720 hours) · injury=yes (chemical) |
| **Replicates** | 2 (this report aggregates 2 successful archive(s); plots show each replicate plus the across-replicate mean) |

## How Replicates Appear in the OMEX

Each replicate is packaged as its OWN per-run .omex (under *replicate_N/*), carrying only that replicate's data and a *replicate_id* / *total_replicates* in run_metadata.json. The per-replicate SED-ML declares a replicate *repeatedTask* over a uniformRange (1..2) [present] — i.e. it documents the N-replicate design even though each archive holds one replicate. This report reads all replicate archives of the combination and aggregates them.

| Replicate | run_metadata | built-in SED-ML | manifest | biosimulators |
|---|---|---|---|---|
| replicate_1 | id 1 of 2 | PASS | PASS | PASS (0 errors, 1 warning group) |
| replicate_2 | id 2 of 2 | PASS | PASS | PASS (0 errors, 1 warning group) |

## Provenance (OMEX Metadata)

| Field | Value |
|---|---|
| Creator (experimenter) | Joel Vanin |
| Created | 2026-10-20T18:20:12 |
| Archived | 2026-10-21T01:01:38 |
| License | https://spdx.org/licenses/CC-BY-4.0.html |
| Citation | Vanin et al. (2025). V-Cornea: a computational model of the corneal epithelium. |
| Keywords | CompuCell3D, cornea, epithelium, agent-based model, wound healing |

From metadata.rdf (COMBINE OMEX-Metadata). The creator is attribution/provenance, not authentication.

## Run Identity (stochastic realization & code)

| Replicate | RNG seed | Origin | Schema fingerprint | Code |
|---|---|---|---|---|
| replicate_1 | 1287481261 | drawn | sha256:d3c5ca23a40e24f9 | build_stamp |
| replicate_2 | 875085774 | drawn | sha256:d3c5ca23a40e24f9 | build_stamp |

Replicates of a combination share a parameterization and differ only in the random number generator seed, so a difference between them is stochastic rather than configured. Seeds marked *drawn* were generated independently at launch and recorded afterwards; seeds marked *explicit* were set by the operator (a pinned seed applies to the first replicate only). The schema fingerprint identifies the parameter definitions in force when the run was configured, and the code entry the source revision that executed it.

## Parameter Credibility (run values against the schema's documented evidence)

79 parameters, 1 changed from default; 27 within documented evidence, 0 extrapolations, 0 inadmissible, 0 numeric without a documented range, 21 switches/categories.

| Parameter | Run value | Default | Unit | Credibility | Evidence for the range assessed |
|---|---|---|---|---|---|
| Initial Chemical Concentration (SLS_Concentration) | 250 | 750 | Arbitrary Units | admissible (no physiological range documented) | calibration: Scott L et al. (2010), Maurer JK et al. (2002), Maurer JK et al. (1997); S10 Table, p. 19 |

*Credibility* places each executed value against the three range tiers the schema documents: the implementation-admissible range (outside = inadmissible), the evidence-supported range (outside = extrapolation beyond the documented evidence) and the recommended range for this model's context of use. The evidence column names the basis and citation the schema records for the supported range; *not recorded* means the bound exists but its justification has not been documented, which is itself an audit finding. Numeric parameters without any documented range are listed as unassessable; switches and categories are not range-assessed. Full detail for every parameter is in parameter_audit.json inside the archive.

## Reading the SED-ML (representative replicate)

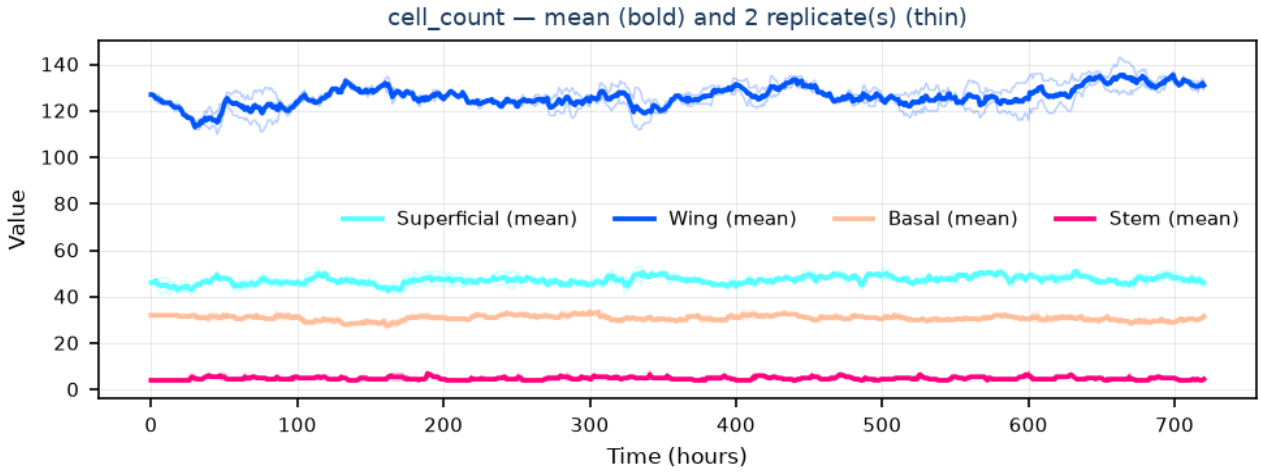
| What the file states | Conclusion |
|---|---|
| **Model** model/vCornea_v2.cc3d · urn:sedml:language:compucell3d | Agent-based CC3D model, **bundled in the archive** so model/@source resolves (self-contained); still descriptive, not re-executable by a SED-ML engine. |
| **Simulation** uniformTimeCourse 0→7200, 7200 steps, KiSAO KISAO:0000241 | 30-day run = 720 h = 7200 MCS (time course is in MCS). |
| **Replicates** repeatedTask present, total_replicates=2 | Stochastic replicates; means/spread across replicates are meaningful (this report uses 2). |
| **DataGenerators**: 11; dataDescriptions: 3 | Signals the SED-ML indexes; thickness parquet is packaged but not indexed. |

## Recorded Outputs (DataDescriptions → columns)

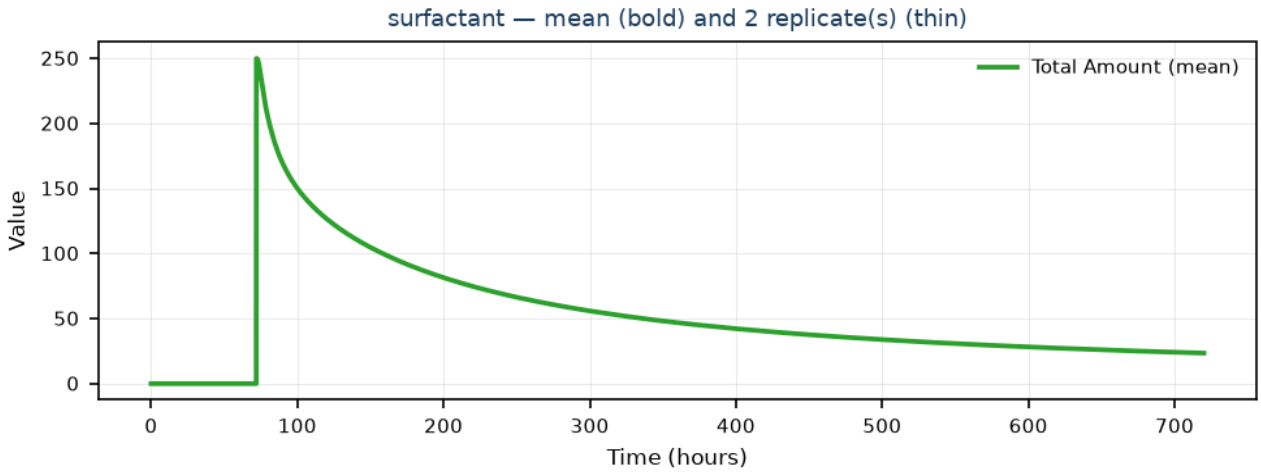
| Source file | Format | Columns |
|---|---|---|
| cell_count_7201.csv | csv | Superficial, Wing, Basal, Stem, Time |
| surfactant_7201.csv | csv | Total Amount, Time |
| thickness_rep_7201.parquet | //purl.org/NET/mediatypes/application/x-parquet | Time, Bin, Height |

## Results (replicate-aware, from the packaged data)

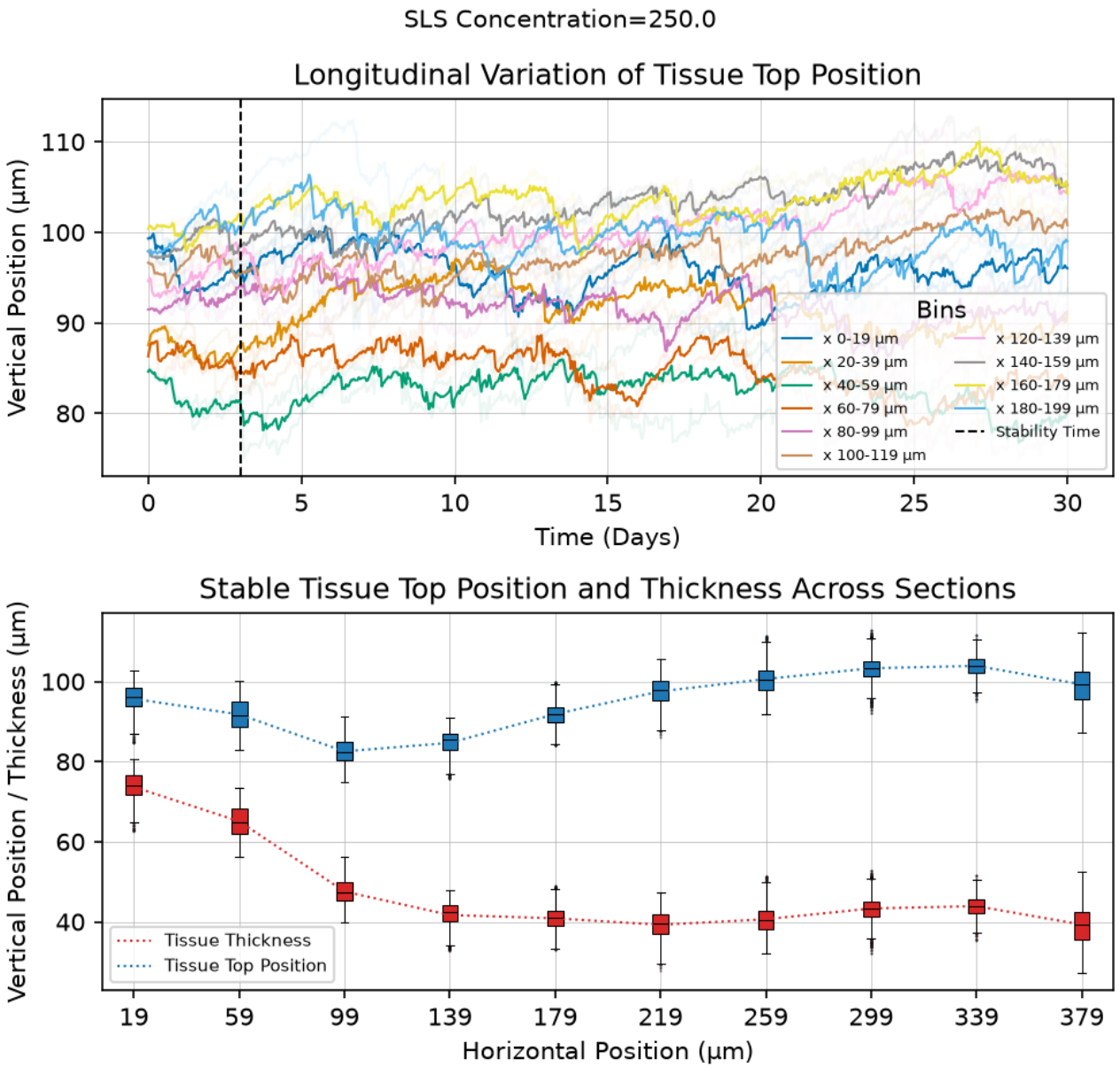
Plotted from the CSV/parquet inside the 2 replicate archive(s). Thin lines are individual replicates; the bold line is the across-replicate mean. Time is shown in hours (as stored in the data).



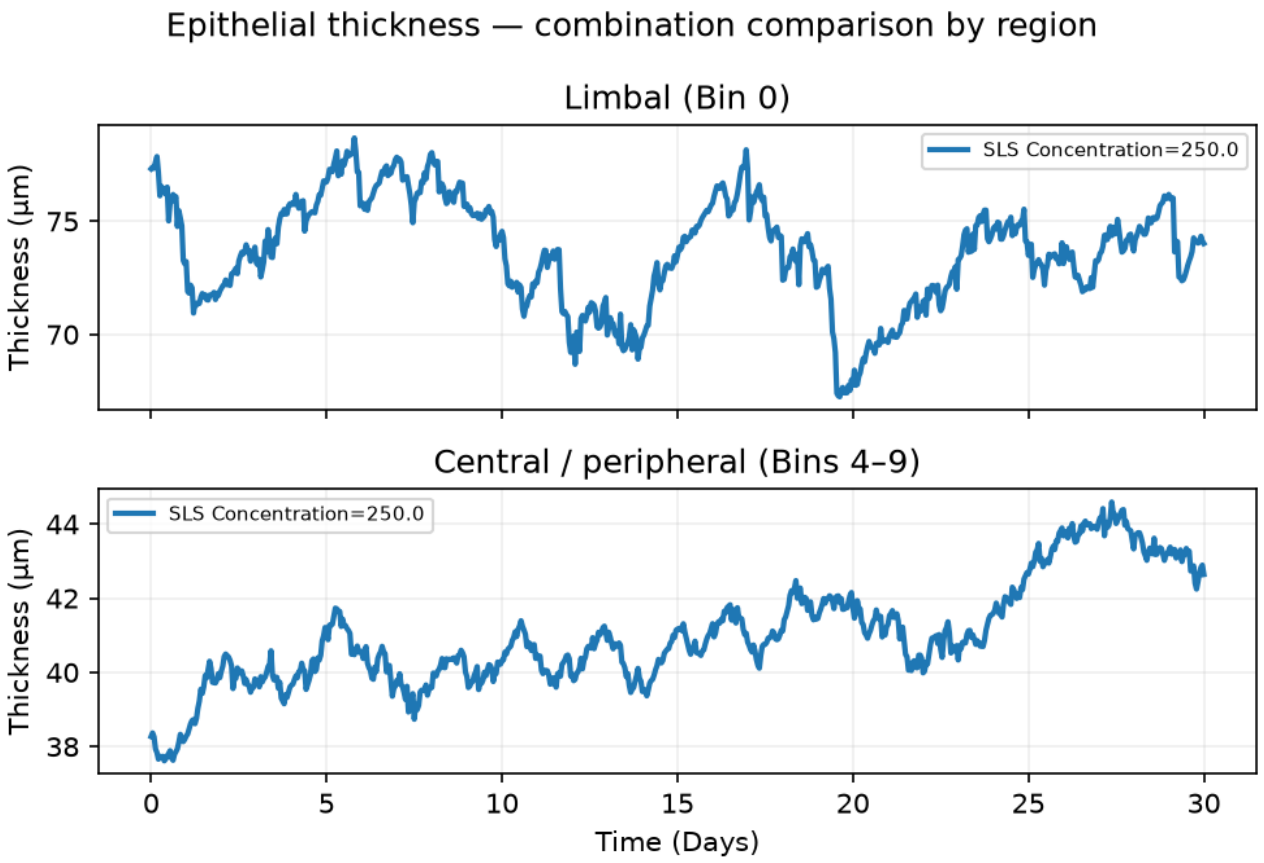

cell_count: Superficial, Wing, Basal, Stem vs Time — 2 replicate(s), bold = across-replicate mean




surfactant: Total Amount vs Time — 2 replicate(s), bold = across-replicate mean




Tissue thickness, drawn by the V-Cornea GUI's own analysis (Analyze Results → Thickness, paper view; Vanin et al. 2025, Fig 8). Top: tissue top position (vertical centre of mass of the top cells) in each of the 10 X-bins over time, as the model records it; faint lines are the 2 replicate(s), solid lines their mean. Bottom: from the stability time (day 3) on, top position (blue) and thickness (red) across X; thickness = (top position – membrane baseline) × 2 µm/voxel.




Epithelial thickness in µm over time in the limbus (bin 0) and the central/peripheral cornea (bins 4–9), averaged over replicates (Analyze Results → Thickness, comparison view).

## Archive Contents (per replicate)

| File | Size (B) | Role |
|---|---|---|
| Epithelium.piff | 692,401 | — |
| Epithelium_new_5545mcs.piff | 656,912 | — |
| Parameters.py | 2,269 | CC3D parameter file applied to this run |
| cell_count_7201.csv | 13,623 | — |
| environment.json | 1,292 | — |
| manifest.xml | 2,519 | OMEX manifest (COMBINE archive index) |
| metadata.rdf | 1,656 | OMEX Metadata (provenance: creator, license, citation) |
| model/Epithelium.piff | 692,401 | Bundled CC3D model file (so model/@source resolves) |
| model/Epithelium_new_5545mcs.piff | 656,912 | Bundled CC3D model file (so model/@source resolves) |
| model/Simulation/Parameters.py | 2,269 | CC3D parameter file applied to this run |
| model/Simulation/vCornea_v2.py | 1,205 | Bundled CC3D model file (so model/@source resolves) |
| model/Simulation/vCornea_v2.xml | 13,788 | Bundled CC3D model file (so model/@source resolves) |
| model/Simulation/vCornea_v2Steppables.py | 67,455 | Bundled CC3D model file (so model/@source resolves) |
| model/vCornea_v2.cc3d | 267 | Bundled CC3D model file (so model/@source resolves) |
| parameter_audit.json | 256,576 | — |
| run_metadata.json | 3,555 | Authoritative parameter/run provenance |
| simulation.sedml | 12,506 | SED-ML L1V4 simulation description |
| status.json | 492 | — |
| stderr.log | 422 | CC3D run log (stderr) |
| stdout.log | 1,422 | CC3D run log (stdout) |

| | | |
|---|---|---|
| surfactant_7201.csv | 170,759 | — |
| thickness_rep_7201.parquet | 76,847 | — |
| vCornea_v2.cc3d | 267 | — |

Each of the 2 replicate archives has this same layout.